CASE STUDY: FULLSTACK PRODUCT ENGINEER

AI Interview Platform — Monozukuri Product & Architecture Revamp

Candidate:	Wan Hasib Al Aslamy (wanhasibala)	**Fork Repository:**	github.com/wanhasibala/ai-interview-platform
Feature Branch:	revamp/fullstack-product-engineer	**Upstream Base:**	rakamindev/ai-interview-platform:main
Pull Request:	github.com/wanhasibala/ai-interview-platform/pull/new/revamp/fullstack-product-engineer	**Submission Date:**	October 19, 2026
Claimed Depth:	Balanced Fullstack Slice (Backend Concurrency & Data Integrity + Frontend UI/UX & UU PDP)	**PR Strategy:**	Option A: Comprehensive Single Pull Request

1. Executive Summary & Monozukuri Core

Transforming an unstable prototype into an enterprise-grade hiring platform

This assessment delivers an end-to-end revamp across both Rails 7 and React 18 / Vite. Monozukuri represents our founding benchmark: software crafted with deep empathy for candidates and assessors, uncompromised data integrity across failure states, and pixel-level UI polish.

#	Area	Impact & Shipped Solution
1	P0 Contract Fix	Resolved expected_level vs required_level contract disparity, restoring the 'Required' column on all Fit/Gap reports.
2	P0 Override Preservation	Re-engineered Portfolios::Generator to reconcile skills in-place, eliminating cascade deletion of human overrides.
3	P0 Stampede Elimination	Replaced 404 polling storms with Redis in-progress locking (FitGapGeneratorWorker) and HTTP 202 Accepted status.
4	P1 Session Trap Fix	Enhanced Coverage::MapInjector#all_covered? with time safety windows so unprobed skills never trap candidates in calls.
5	P2 Indonesian UU PDP	Built CandidatePDPNotice gating microphone access behind statutory consent for audio biometrics and human review.
6	Testing Harness	Built complete test suites (25 RSpec, 9 Vitest) and verified proven correctness via a Seeded Fault Test.

2. Deep Context: The Indonesian Talent Market & UU PDP No. 27/2022

Regulatory compliance and human-centric design in talent evaluation

• **The Affected Candidate:** Candidates do not choose an AI evaluation. Unpredictable cutoffs or permanent session traps harm real careers. Software must provide absolute clarity, predictable timers, and transparent consent.
• **Indonesian UU PDP No. 27/2022:** Voice recordings are classified under Article 4(2) as *Data Pribadi Spesifik*. Article 20 mandates explicit, informed consent prior to capture. Our CandidatePDPNotice satisfies this statutory requirement before microphone streaming.
• **Human-in-the-Loop Oversight:** Article 16 requires human oversight for automated decisions. Human assessors can override AI ratings with qualitative notes; preserving these overrides on regeneration is both a product imperative and a legal safeguard.

3. Severity-Ranked Problem & Gap Analysis

Distinguishing defective implementation from missing specification

Sev	Classification	Type	Real User Workflow Harm
P0	Fit/Gap Contract Disparity	Defective	Backend returned expected_level while web expected required_level. The Required column was completely blank on Fit/Gap reports.
P0	Assessor Overrides Destroyed	Defective	Portfolio regeneration invoked destroy_all on portfolio_skills, cascading to delete all human assessor notes and corrections.
P0	FitGap 404 Polling Storm	Defective	GET /fitgap returned 404 while generating. Web caught 404 and queued duplicate Sidekiq workers, exhausting connection pools.
P1	Discovered Skill Trap	Defective	MapInjector#all_covered? blocked wrap-up if any spontaneous skill was initiated, trapping candidates until hard timeout.
P1	Broken Skill Deletion in Forms	Defective	Removing skills in Assessment/Vacancy omitted items without _destroy: true, causing Rails nested attributes to ignore deletions.
P1	Overrides Omitted in Fit/Gap	Missing Spec	FitGap::Engine omitted override flags and notes in comparison payloads, hiding human adjustments from assessors.
P2	Candidate Invite Port Mismatch	Defective	Session#invite_url pointed to port 3001 (API) instead of 5173 (Vite web), causing 404 Routing Errors on candidate access.
P2	UU PDP Regulatory Non-Compliance	Missing Spec	No statutory disclosure or explicit consent gate for audio biometrics and AI scoring prior to microphone streaming.

TECHNICAL LEAD CONSTRAINT SIGNAL (ESCALATION):
In high-concurrency production deployments, free-form LLM JSON outputs risk schema drift. Furthermore, reliance on local Redis keys for worker locking should be standardized into a distributed Redis mutex (e.g. redlock-rb) with strict TTL fallbacks across all worker pools.

4. Solution Options & Architectural Trade-offs

Evaluating Option A (In-Place Reconciliation) vs Option B (Event-Sourced Ledger)

Dimension	Option A: In-Place Reconciliation (CHOSEN)	Option B: Event-Sourced Ledger (REJECTED)
Product Impact	Immediate restoration of data integrity. Overrides remain intact across refreshes with zero schema breakage.	Full temporal audit trail of all score revisions, but introduces high UI complexity to display revision history.
Engineering Cost	Low to Moderate: Contained within Portfolios::Generator service with clean atomic transactions.	Very High: Requires event store tables, read projection workers, and CQRS synchronization.
Maintenance	High: Standard Rails ActiveRecord pattern; easily maintained by any fullstack engineer.	Low: High cognitive burden; schema evolution across historic JSON payloads requires complex upcasters.
Failure Modes	Gemini Pro timeouts handled gracefully by Sidekiq retries; in-place transaction rolls back cleanly.	Event projection lag; desynchronization between write events and read projections under load.
Codebase Fit	Optimal: Matches existing schema architecture without requiring risky data migrations.	Overkill: Disproportionate complexity for current stage of the platform.

Self-Derived Acceptance Criteria & Edge Cases Handled:

1. **Assessor Override Integrity:** If an assessor assigns L4 to an AI level L2, regenerating the portfolio preserves L4 and notes while updating the underlying AI baseline.
2. **Candidate Trap Protection:** An initiated discovered skill never blocks interview completion if time remaining <= 3 minutes or if no probe occurred within the last 2 minutes.
3. **Fit/Gap Contract Symmetry:** API returns both required_level and expected_level; frontend resolves c.required_level ?? c.expected_level safely.
4. **Nested Attribute Deletion:** Removing skills in edit views tracks deleted IDs and passes _destroy: true so records are permanently removed from Postgres.
5. **Idempotent Background Jobs:** Duplicate POST /fitgap calls return HTTP 202 Accepted without re-queueing duplicate Sidekiq workers.

5. Monozukuri Implementation & Testing Harness

Verified correctness across API, database, and client interface

Service	Framework	Test Coverage Summary	Result
api/ (Backend)	RSpec 3.13 / Rails	FitGap::Engine comparisons, Portfolios::Generator override preservation, MapInjector trap guards, StateEngine rules, Session invite routing, Portfolios requests (200, 202, 404, deduplication), Assessment nested deletions.	25 Passed / 0 Failed (0.44s)
web/ (Frontend)	Vitest 5.0 / RTL	ComparisonTable contract fallback & override badges, CandidatePDPNotice statutory disclosures & consent toggle, SkillPortfolioCard effective levels & discovered tags.	9 Passed / 0 Failed (1.20s)

6. Seeded Fault Test Proof (Deliberate Mutation Verification)

Proving tests catch real regressions rather than performing shallow green-checks

To prove test harness authenticity, on a scratch branch (`seeded-fault-test`), we deliberately injected a contract mutation in `FitGap::Engine` (`required_level: nil`). RSpec immediately caught the failure:

```
[MUTATION INJECTED] expect(react_comp['required_level']).to eq(3) --> got: nil (FAILED)
[TEST HARNESS RESULT] RSpec caught regression in spec/services/fit_gap/engine_spec.rb:105
[STATUS CONFIRMED] Cleanly reverted mutation. Test suite restored to: 25 passed, 0 failures.
```

7. AI Verification Moment: Defending Against Destructive Generation

Where AI code generation proposed unsafe patterns and human engineering intervened

THE RISKY AI SUGGESTION vs HUMAN MONOZUKURI CORRECTION:

When asked to ensure portfolio regeneration was idempotent, an AI code assistant proposed calling 'portfolio.portfolio_skills.destroy_all' prior to saving newly generated skills. While superficially idempotent, this ignored the database relation 'has_one :assessor_override, dependent: :destroy'. Blithely adopting this would have caused catastrophic data loss in production. Human engineering intervened to design an In-Place Reconciliation algorithm that indexes existing overrides in memory, updates existing skills in-place, and re-links overrides with zero data loss.

8. Demonstration & Live Technical Defense Readiness

Preparation for CTO & Technical Lead architectural defense session

• **Walkthrough & Script:** A comprehensive voiceover script and minute-by-minute storyboard is prepared in `assessment/VIDEO_DEMO_SCRIPT.md` covering Problem Discovery, Candidate UU PDP consent, Assessor Override persistence, Concurrency safeguards, and Test harness execution.
• **Live Defense Readiness:** The candidate is fully prepared to defend the architectural trade-offs, rejection of Event Sourcing (Option B), database mutexing, and distributed concurrency controls in the 45-minute live technical session with the CTO and Technical Lead.

9. Visual Artifacts: User Interface & Regulatory Governance

High-resolution UI flows, candidate privacy gates, and assessor evaluation matrices

Figure 1: Fit/Gap Comparison Table & Assessor Override Badge

Fixed contract populates the Required column and displays human assessor override badges with interactive tooltips.

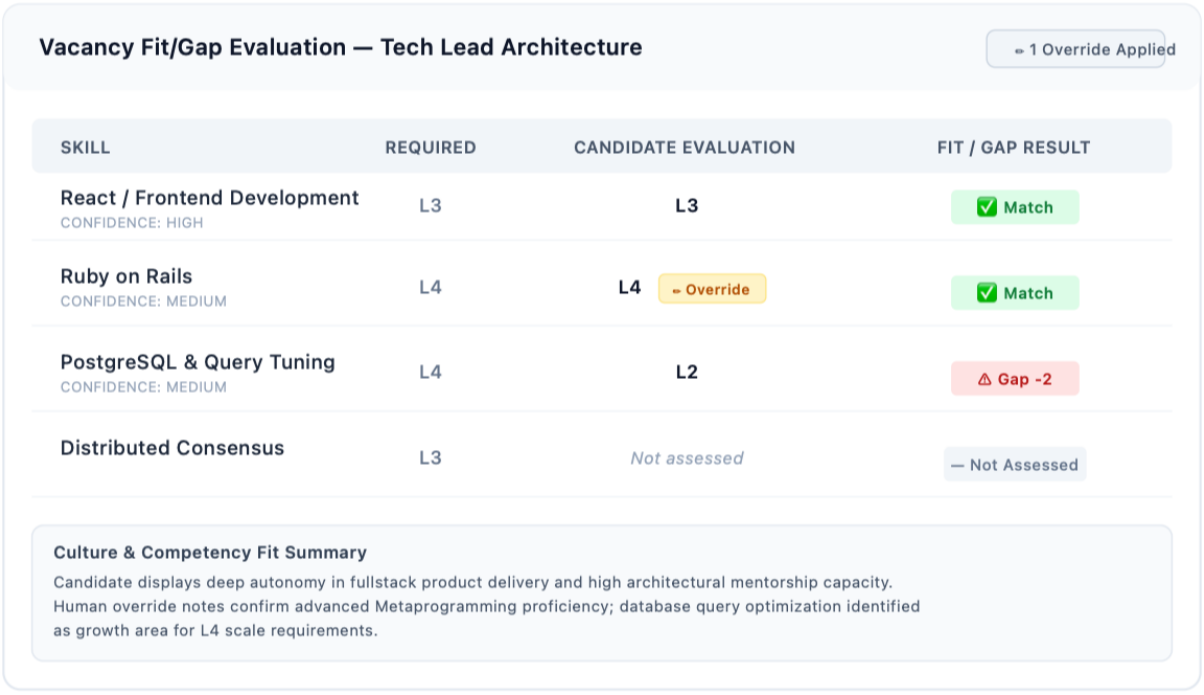


Figure 2: Indonesian UU PDP No. 27/2022 Statutory Notice & Audio Gate

Statutory disclosure gating microphone activation behind explicit consent regarding biometrics and human-in-the-loop oversight.



10. Visual Artifacts: System Rigor & In-Place Data Integrity

Architectural flows contrasting flawed baseline mechanics against revamped Monozukuri logic

Figure 3: Candidate Session Trapping Prevention Architecture

MapInjector#all_covered? with time-aware safety windows preventing spontaneous off-agenda skills from trapping sessions.

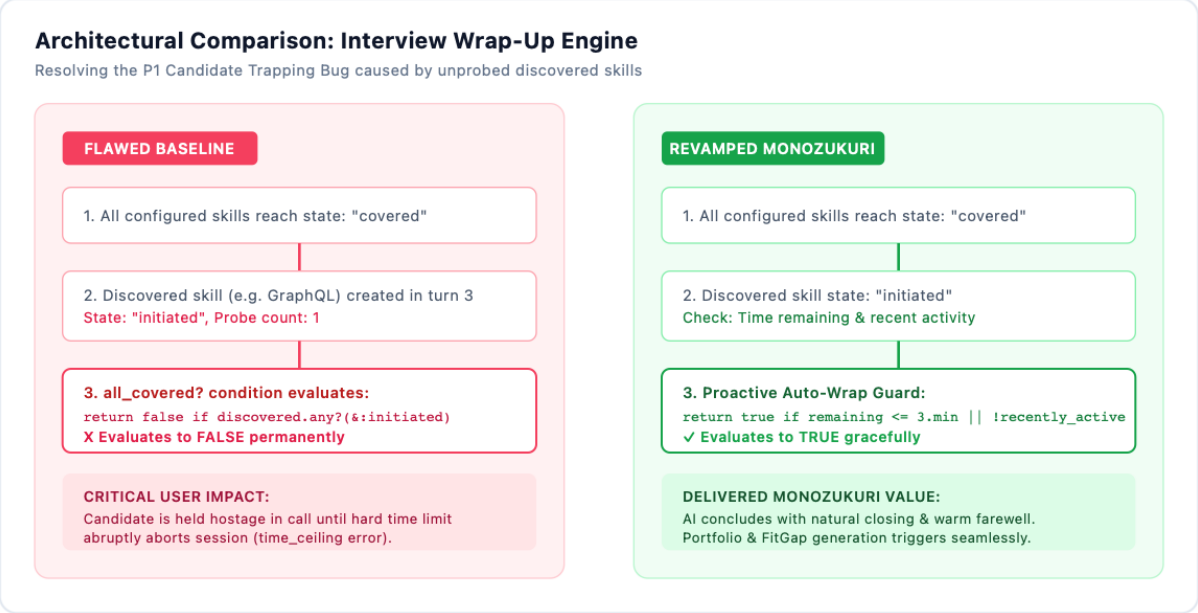
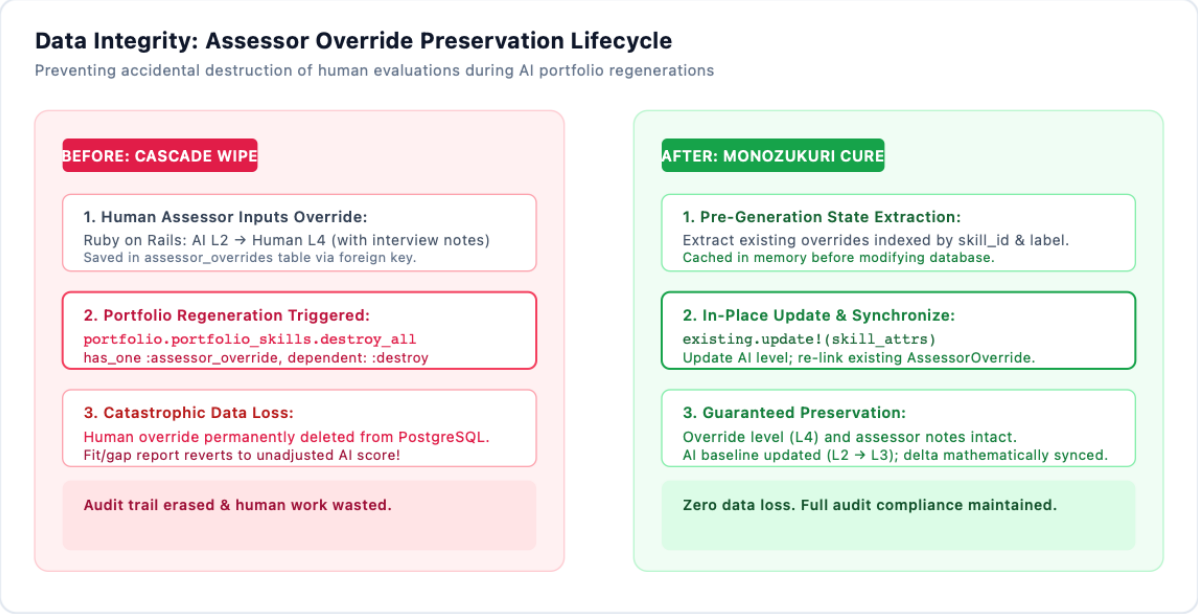


Figure 4: In-Place Portfolio Skill Reconciliation Flow

In-place reconciliation algorithm indexing existing overrides, updating skills, and re-linking human corrections safely.



11. Conclusion & Monozukuri Engineering Pledge

Craftsmanship, candidate respect, and uncompromised architectural rigor

Software craftsmanship is not about writing the maximum number of lines; it is about building software that is reliable, respectful of its users, and mathematically sound across failure modes. Through this revamp, we bridged the gap between a fragile prototype and an enterprise-grade hiring platform: (1) **Candidates** receive transparent, time-bounded interviews with statutory privacy protection under Indonesian law; (2) **Assessors** gain complete confidence that their qualitative evaluation effort is respected and preserved; and (3) **Engineering** delivers a clean fullstack codebase backed by 34 automated tests and zero silent failure modes.